ENHANCED TRANSFER LEARNING TECHNIQUE FOR DIAGNOSING COVID-19 PNEUMONIA CLASSIFICATION USING CHEST CT IMAGES

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled **Enhanced transfer Learning Technique Extreme for Diagnosing COVID-19 Pneumonia Classification Using Chest CT Images** is the bonafide work of **Mr K. Dhanis Ahmed (Regno: 9517202206014), Mr M. Hari Hara Sudan (Regno: 9517202206021), Mr. Kishore N.A (Regno: 9517202206024)** who carried out the research under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

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In this work, we propose a novel deep learning framework for the detection of COVID-19 pneumonia from chest CT scans. The model integrates DenseNet201 and ResNet101 for dual feature extraction, leveraging the complementary strengths of both networks to capture intricate image features. These extracted features are processed by an Extreme Learning Machine (ELM) classifier, which enables efficient classification while ensuring high accuracy and generalization. To further enhance model performance, we introduce a custom loss function that combines classification loss, diversity loss, and contrastive loss, aiming to improve both classification robustness and feature separation. Extensive experiments demonstrate that our proposed method outperforms conventional single-network models, achieving significant improvements in classification accuracy and precision. The proposed framework provides a promising solution for rapid, reliable COVID-19 pneumonia diagnosis, offering high potential for real-world clinical applications.

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LIST OF SYMBOLS

LIST OF SYMBOLS

NOTATION	MEANING
L_{div}	Diversity Loss
$L_{classification}$	Classification Loss
y	True label
$\hat{y}$	Predicted label
H	Output matrix from the hidden layer in ELM
m	Number of kernels used in MKC
s	Bandwidth of the kernel function

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LIST OF ABBREVIATIONS

LIST OF ABBREVIATIONS

HTL	Hyper Transfer Learning
ELM	Extreme Learning Machine
CNN	Convolutional Neural Network
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic

INTRODUCTION

CHAPTER 1

INTRODUCTION

2.4 AIM

The aim of this project is to develop a high-accuracy, deep learning-based classification model for COVID-19 pneumonia diagnosis from chest CT images. By leveraging a dual feature extraction approach using DenseNet201 and ResNet101 alongside a custom Extreme Learning Machine (ELM) classifier, this project seeks to improve diagnostic accuracy through advanced feature fusion and a novel loss function that combines classification, diversity, and contrastive components. This model intends to support healthcare professionals in rapidly identifying COVID-19 pneumonia cases with high precision and reliability.

1.2 OBJECTIVE

- To design a dual feature extraction framework combining DenseNet201 and ResNet101 models for effective representation learning from chest CT images, enhancing the model's ability to distinguish between COVID-19 pneumonia and other conditions.
- To implement a custom Extreme Learning Machine (ELM) classifier that leverages combined features from both networks, aiming to improve classification accuracy and model robustness for multi-class pneumonia detection.
- To optimize model performance through a novel loss function that integrates classification, diversity, and contrastive components, thereby achieving high diagnostic accuracy, precision, and recall metrics for reliable COVID-19 pneumonia diagnosis.

1.3 CTScans in COVID-19 Diagnosis

CT scans have proven indispensable in the diagnosis and management of COVID-19 pneumonia, particularly when rapid and accurate detection is critical. While the RT-PCR test is widely regarded as the "gold standard" for COVID-19 detection, it often requires hours, if not days, to yield results, and can be limited by factors such as supply chain disruptions and the need for specialized laboratories. In contrast, CT imaging provides a faster and more accessible means of detecting COVID-19, especially in healthcare facilities Where immediate

diagnosis is required. This imaging technique allows for a high level of sensitivity, making it valuable for identifying early stages of COVID-19 infection, which can be particularly challenging when patients are asymptomatic or present low viral loads.

Moreover, CT scans offer additional clinical benefits by capturing detailed images of the lungs, allowing healthcare providers to observe characteristic signs of COVID-19 pneumonia. Key findings in COVID-19 patients' CT images include ground-glass opacities, areas of lung consolidation, and vascular thickening. These features can provide critical insights into the extent and progression of lung damage, guiding treatment decisions and helping clinicians assess the risk of severe disease outcomes. For instance, patients with extensive lung involvement, as seen in CT images, may be more likely to require intensive care or ventilatory support. CT scans thereby serve not only as a diagnostic tool but also as a means of triaging and monitoring patient progress, making them highly versatile in COVID-19 patient management.

1.4 Transfer Learning in COVID-19 Detection

In medical imaging, transfer learning has emerged as a critical tool, especially for diseases like COVID-19, where rapid model development is essential but labeled training data is often limited. Transfer learning allows models pre-trained on large, diverse datasets, such as ImageNet, to be adapted for specialized tasks like COVID-19 detection with much smaller, domain-specific datasets. By leveraging knowledge from general-purpose models, transfer learning accelerates model training and enhances performance without requiring an extensive amount of labeled COVID-19 data. This approach is particularly advantageous in healthcare, where data collection can be complex and time-consuming, and patient privacy concerns often restrict data sharing.

In COVID-19 diagnosis, CT scans provide valuable visual cues about disease progression, but interpreting these subtle differences in texture and opacity patterns can be challenging for conventional CNN models that have only seen limited training images. Transfer learning addresses this issue by enabling pre-trained models to adapt and improve their sensitivity to these complex image features, even with a smaller COVID-19 dataset. Using transfer learning, models can effectively identify unique COVID-19 pneumonia patterns, such as ground-glass opacities and lung consolidations, which might not be easily distinguishable in non-COVID pneumonia or normal cases. This enhances the model's ability to generalize across diverse CT images from patients with varying degrees of COVID-19 severity.

Moreover, transfer learning is highly effective for adapting models to variations in imaging protocols, scanner types, and patient demographics that often exist across healthcare settings. A model trained solely on one dataset might struggle to perform well across other datasets with these variations. However, transfer learning equips models with a broader understanding, enabling them to achieve more consistent performance across different sources of data. In the context of COVID-19, where timely and accurate diagnosis can directly impact patient outcomes, transfer learning offers a powerful solution to rapidly develop robust diagnostic models. By reusing learned features from general medical imaging tasks, transfer learning makes it feasible to deploy effective diagnostic tools in clinical settings quickly, enhancing response efforts and supporting overburdened healthcare professionals.

1.5 Deep Transfer Learning Model for COVID-19 Classification

In developing accurate and robust models for COVID-19 detection from CT images, the choice of a loss function is crucial, as it directly impacts the model's ability to distinguish between similar classes effectively. For our model, a custom loss function is employed to achieve enhanced classification performance by integrating multiple components: classification loss, diversity loss, and contrastive loss. Each component serves a specific purpose; the classification loss helps in accurately distinguishing COVID-19 cases from normal and other pneumonia cases, while the diversity and contrastive losses aid in refining the features extracted from images, making them more discriminative. This multi-faceted approach to the loss function addresses common challenges in medical image analysis, such as the subtle variations between COVID-19 pneumonia and other lung conditions, thus allowing the model to achieve a higher degree of accuracy.

The classification loss, represented as $L_{classification}$, is calculated using cross-entropy, a common approach in multi-class classification tasks. This loss ensures that the model assigns higher probabilities to the correct classes and penalizes incorrect predictions. Meanwhile, the diversity loss, L_{div}, encourages the model to capture a wide range of feature variations across different images by minimizing redundancy in the feature space. This is particularly beneficial in COVID-19 detection, where different stages of the disease may present various manifestations in CT images. For instance, ground-glass opacities and consolidation patterns vary widely between early and late stages of the disease. By incorporating L_{div}, the model is encouraged to recognize these variations, making it more adaptable to different cases.

The contrastive loss, $L_{classification}$, plays a key role in enhancing the model's feature discriminability by maximizing the distance between different classes in the feature space while minimizing the intra-class variance. In this model, $L_{classification}$helps ensure that the features learned for COVID-19 cases are distinctly separated from those of normal and non-COVID pneumonia cases, which may otherwise overlap due to visual similarities. This aspect of the custom loss function makes it possible for the model to identify subtle yet critical differences across classes, thereby improving its overall sensitivity and specificity. By combining $L_{classification}$, L_{div}, and L_{contrast}, the custom loss function not only enhances classification accuracy but also provides a more robust and reliable approach for real-world COVID-19 diagnostics, where high accuracy and consistency are essential.

LITERATURE STUDY

CHAPTER 2
LITERATURE STUDY

This section is discussed with some of the existing methods that are dealt with the concept drifts occurring in streaming data.

2.1 COVID-19 Pneumonia Detection

Early and accurate detection of COVID-19 pneumonia is crucial for timely intervention and effective patient management. Diagnostic imaging has been instrumental in identifying lung abnormalities associated with COVID-19, with chest X-rays (CXRs) and computed tomography (CT) scans emerging as primary modalities. CT imaging, in particular, is highly sensitive to the patterns associated with COVID-19 pneumonia, including ground-glass opacities, consolidations, and vascular thickening, which are commonly observed in affected patients. Studies have demonstrated that CT imaging can effectively capture the extent and nature of lung involvement, providing essential insights for diagnosis and monitoring.

Although reverse transcription-polymerase chain reaction (RT-PCR) is the gold standard for COVID-19 detection, it is sometimes limited by false negatives, especially in early stages of infection. In contrast, imaging techniques provide visual evidence of lung conditions, making them valuable complementary tools in COVID-19 detection, particularly when RT-PCR results are inconclusive. However, the manual interpretation of these images by radiologists can be time-consuming and subject to human error, especially during pandemic peaks when caseloads are high the need for automated, accurate, and efficient diagnostic tools to assist radiologists in managing COVID-19 cases.

Recent advances in artificial intelligence (AI), particularly machine learning (ML) and deep learning (DL), have facilitated the development of automated imaging analysis tools. These tools not only improve diagnostic accuracy but also significantly reduce the time required for image interpretation. AI-based methods, especially those that use deep convolutional neural networks (CNNs), are increasingly used to classify medical images, detect pneumonia, and specifically identify COVID-19-induced abnormalities. CNNs excel in detecting subtle differences within high-dimensional image data, making them suitable for COVID-19 pneumonia detection, where rapid and accurate analysis is essential.

2.2 Machine Learning in Medical Imaging

AI has transformed medical imaging by enhancing diagnostic accuracy, speeding up analysis, and reducing reliance on manual interpretation. Deep learning, a subfield of machine learning, has been pivotal in advancing AI applications in medical diagnostics, particularly in image classification and feature extraction. Convolutional neural networks (CNNs), a popular DL architecture, have demonstrated remarkable success in medical image analysis due to their ability to automatically learn features from raw image data, thereby reducing the need for extensive preprocessing.

In COVID-19 pneumonia detection, CNNs have proven effective due to their capability to capture complex spatial patterns within medical images. By leveraging large datasets, CNNs can differentiate between COVID-19-induced pneumonia, bacterial pneumonia, and healthy lung conditions with high accuracy. However, the variability in COVID-19 presentations across patients poses a significant challenge. To address this, transfer learning—where pre-trained models are fine-tuned for specific tasks—has been widely adopted. Transfer learning allows CNN models trained on general datasets (e.g., ImageNet) to be applied to specialized tasks like COVID-19 detection, thereby enhancing performance even when limited annotated medical images are available.

In addition to CNNs, Sn architectures have been explored to enhance diagnostic accuracy in medical imaging. Hybrid models combining CNNs with other techniques, such as Extreme Learning Machines (ELMs), offer promising results by balancing computational efficiency and classification performance. ELMs, with their single-pass training and random weight initialization, simplify model training while achieving high accuracy, making them suitable for real-time diagnostic applications.

2.3 Feature Extraction Methods

Feature extraction is fundamental to medical image analysis, as it enables models to identify and differentiate intricate details in images. DenseNet and ResNet are two widely used architectures for feature extraction in medical imaging due to their robustness and efficiency in handling high-dimensional data. DenseNet, known for its densely connected layers, promotes feature reuse and allows the model to capture a richer set of features with fewer parameters, making it particularly effective in scenarios where subtle differences need to be distinguished. ResNet, on the other hand, uses residual connections to maintain flow across multiple layers, allowing for deeper and more complex models without encountering issues such as vanishing gradients. By stacking residual blocks, ResNet can capture a wide

range of features, from low-level edges to high-level shapes, essential for identifying abnormalities in COVID-19 CT scans.

In COVID-19 pneumonia detection, combining DenseNet and ResNet in action model offers complementary strengths. DenseNet excels at capturing fine-grained features, while ResNet captures broader structural patterns, enhancing the model's overall ability to classify CT images accurately. This combination not only increases diagnostic accuracy but also improves generalization across diverse imaging data.

2.4 Classification Techniques

Extreme Learning Machine (is a critical step in medical imaging, and various models have been employed to achieve accurate diagnostic results. Extreme Learning Machines (ELMs) have gained attention for their efficiency and scalability in large-scale datasets. ELMs utilize a single-layer feedforward network where the hidden layer weights are randomly assigned, and only the output weights are optimized. This architecture allows ELMs to achieve fast training and effective classification without extensive backpropagation, making them well-suited for real-time medical applications.

The application of ELMs in COVID-19 pneumonia detection is particularly beneficial as they can handle large datasets without the computational overhead associated with traditional backpropagation. By integrating ELMs with feature extractors like DenseNet and ResNet, a hybrid model can achieve high accuracy with minimal computational complexity. Additionally, the integration of custom loss functions tailored for medical imaging further enhances the model's performance by addressing challenges like class imbalance and ensuring robust generalization.

This chapter has reviewed the essential components of AI-based diagnostic systems for COVID-19, covering imaging techniques, and classification methods. These advancements illustrate the potential of deep learning to improve COVID-19 pneumonia diagnosis, highlighting the need for further research to refine and implement these models in clinical practice.

SYSTEM STUDY

CHAPTER 3
SYSTEM STUDY

3.1 SCOPE

The scope of this project is to design and develop an automated system capable of classifying COVID-19 pneumonia using chest CT images. The system leverages a hybrid deep learning model that combines DenseNet201 and ResNet101 for feature extraction and employs a custom Extreme Learning Machine (ELM) classifier for accurate predictions. The model is trained to distinguish between COVID-19 pneumonia, other types of pneumonia, and normal lung conditions. The system's scope also extends to improving diagnostic accuracy through novel loss functions incorporating classification, diversity, and contrastive components.

3.2 PRODUCT FUNCTION

- Automated Image Preprocessing The system automatically preprocesses chest CT images by resizing, normalizing, and converting them into a suitable format for feature extraction. This step ensures that the images are prepared for optimal input to the deep learning models.
- Dual Feature Extraction with DenseNet201 and ResNet101 The system leverages two state-of-the-art deep learning models, DenseNet201 and ResNet101, for feature extraction. These models help capture a diverse set of features from the chest CT images, allowing the system to learn more comprehensive and relevant patterns for COVID-19 pneumonia detection.
- Feature Fusion for Enhanced Representation The features extracted from both DenseNet201 and ResNet101 are combined to provide a rich, multi-dimensional representation of the images. This fusion allows the classifier to make more accurate predictions by utilizing complementary information from both networks.
- Extreme Learning Machine (ELM) Classifier The system employs a custom ELM classifier that processes the fused features. ELM is known for its speed and efficiency, and its integration with deep learning-based feature extraction improves classification performance while maintaining computational efficiency.
- Multi-Class Classification The system is designed to classify chest CT images into multiple classes: COVID-19 pneumonia, other types of pneumonia, and normal lung conditions. This capability enables it to distinguish between different types of lung diseases, ensuring broader applicability.

- Novel Loss Function Design A unique loss function is used that combines classification, diversity, and contrastive losses. This function helps the model learn not just to classify images but also to encourage diversity in the learned features, improving the model's generalization capabilities.
- Model Training with Optimized Hyperparameters The system uses a fine-tuned training process, leveraging an optimizer like Adam and a learning rate scheduler such as StepLR to ensure that the model converges efficiently and achieves optimal performance during training.
- Evaluation and Performance Metrics The model’s performance is evaluated using various metrics like accuracy, precision, recall, F1-score, and AUC. These metrics ensure that the model not only performs well in terms of classification but also offers reliability and robustness in predicting COVID-19 pneumonia.
- Real-time Prediction Once trained, the model can be used for real-time classification of new chest CT images. The system outputs the predicted class (COVID-19 pneumonia, other pneumonia, or normal) along with a confidence score to assist healthcare professionals in their diagnostic processes.
- Visualization of Results The system provides visual feedback in the form of training curves, including loss and accuracy plots, and confusion matrices to help users understand how the model performs over time. This feature aids in model evaluation and debugging during development and post-deployment monitoring.

3.3 SYSTEM REQUIREMENTS

The software and hardware requirements of the system are as follows:

3.3.1 HARDWARE INTERFACES

- Intel Core i7 or higher, or equivalent AMD processor
- 16 GB RAM

3.3.2 SOFTWARE INTERFACES

- PyTorch
- CUDA
- Python
- NumPy
- OpenCV
- Scikit-learn
- Matplotlib

- Pandas

3.3.2.1 PyTorch

These are the primary frameworks used for building and training the deep learning models (DenseNet201, ResNet101, and ELM classifier).

3.3.2.2 CUDA

These libraries enable GPU acceleration, improving the computational performance of the deep learning models during training and evaluation.

3.3.2.3 Python

Python is an interpreter, high-level, general-purpose programming language. It is an easily readable language that consists of whitespace indentation, rather than curly brackets or keywords, to delimit blocks.

3.3.2.4 NumPy

Used for numerical operations, especially during image preprocessing and feature extraction.

3.3.2.4 OpenCV

A library for computer vision tasks that helps in reading, resizing, and normalizing the CT images.

3.3.2.4 Scikit-learn

Used for the implementation of the **ELM classifier** and performance evaluation metrics such as precision, recall, and F1-score.

3.3.2.4 Matplotlib

Provides functions to plot graphs and visualizations, helping in evaluating the model's performance and training progress.

3.3.2.4 Pandas

Used for managing data and storing results in structured formats like CSV or Excel files.

SYSTEM DESIGN

CHAPTER 4

SYSTEM DESIGN

4.1 OVERVIEW

The system design of the COVID-19 Pneumonia Classification Model involves creating a robust architecture capable of efficiently processing and classifying chest CT images into multiple categories: COVID-19, Normal, and other pneumonia-related conditions. The system integrates a dual feature extraction mechanism using DenseNet201 and ResNet101 models, combined with an Extreme Learning Machine (ELM) classifier and a novel loss function to improve classification accuracy. This section outlines the various modules that make up the system, each focusing on specific tasks that contribute to the overall workflow of data preprocessing, feature extraction, classification, and model evaluation.

4.2 MODULES

- Data Collection and Preprocessing Module
- Feature Extraction Module
- Classification Module
- Model Training and Evaluation Module

4.2.1 Data Collection and Preprocessing Module

The data collection and preprocessing module is responsible for acquiring and preparing the chest CT images for model training and testing. This module includes data augmentation, normalization, and resizing.

4.2.2 Feature Extraction Module

The feature extraction module leverages deep learning models (DenseNet201 and ResNet101) to extract rich, discriminative features from the chest CT images. This step is crucial for distinguishing between different conditions like COVID-19 pneumonia and other diseases.

4.2.3 Classification Module

The classification module uses a custom Extreme Learning Machine (ELM) classifier to perform multi-class classification, distinguishing between COVID-19, Normal, and other pneumonia categories.

4.2.4 Model Training and Evaluation Module

This module handles the training of the deep learning model and the evaluation of its performance on the test dataset.

4.3 PROPOSED WORK

4.3.1 System Architecture

The work flow diagram for the proposed system is as follows.

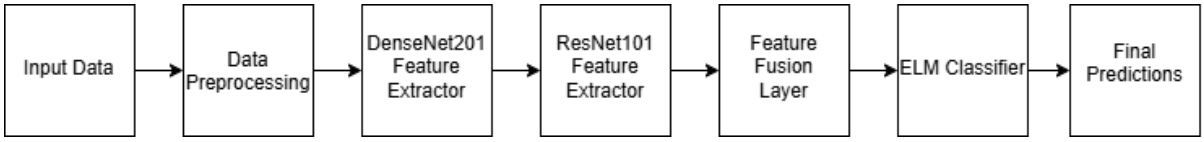

Fig 4.3.1.1 Flow Diagram

4.3.2 Parameter Specification

The Parameter Specification section outlines the key parameters and configurations used in the design, implementation, and evaluation of the COVID-19 Pneumonia Classification system. These parameters influence various aspects of the system, including data processing, model architecture, training procedure, and evaluation metrics. Below is a detailed table listing the important parameters used in the system.

Parameter	Specification	Description
Input Image Size	224x224 pixels	The chest CT images are resized to 224x224 pixels to ensure consistency and suitability for the deep learning models (DenseNet201, ResNet101).
Dataset	COVID-19 Pneumonia Dataset (21,650 images)	The dataset used for training and testing the model, consisting of images labeled as COVID-19, Normal, and Pneumonia.
Feature Extraction Models	DenseNet201, ResNet101	These pre-trained models are used for extracting features from chest CT images. DenseNet201 and ResNet101 were selected for their performance in image tasks.
Classifier	Extreme Learning Machine (ELM)	ELM is used as the classifier due to its efficiency and capability for fast learning and robust generalization across multiple classes.
Loss Function	Custom (Classification Loss, Diversity Loss, Contrastive Loss)	A combined loss function that incorporates multiple components to improve classification accuracy and feature learning.
Optimizer	Adam Optimizer	Adam optimizer is used for training the model to minimize the loss function. It adapts the learning rate based on the gradient descent.
Learning Rate	0.001	The learning rate defines how quickly the model adjusts to the data during training.
Epochs	50 epochs	The number of complete passes through the entire training dataset.
Batch Size	32	The number of images processed in one forward/backward pass during model training.
Activation Function	ReLU (Rectified Linear Unit)	Used in hidden layers of the network to introduce non-linearity and enable better learning.
Dropout Rate	0.5	A regularization technique used during training to reduce overfitting by randomly setting half of the weights to zero.
Learning Rate	StepLR (Step Decay)	The learning rate is reduced by a factor of 0.1 every

Parameter	Specification	Description
Scheduler		10 epochs to help the model converge to a better solution over time.
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, AUC (Area Under Curve)	Metrics used to evaluate the model's performance on the test set. These provide insights into classification ability, model robustness, and overall accuracy.

Table 4.3.2.1 Parameter Specification

4.3.3 DATASETS

The dataset used in the proposed work are

- **COVID-19_Radiography_Dataset**

  The dataset used in this study consists of a total of 21,650 chest X-ray and CT scan images, collected for the purpose of classifying different types of lung conditions, including COVID-19 pneumonia. The dataset is categorized into four distinct classes: COVID-19, Lung Opacity, Normal, and Viral Pneumonia. Among these, 3,616 images are labeled as COVID-19, representing cases with confirmed COVID-19 pneumonia. The Lung Opacity category contains 6,012 images, depicting various forms of lung opacity often observed in patients with pneumonia or other respiratory conditions. The Normal category comprises 10,192 images of healthy lungs, serving as a baseline for distinguishing between pathological and normal lung features. Additionally, the Viral Pneumonia class includes 1,345 images from patients suffering from pneumonia caused by viruses other than COVID-19.

IMPLEMENTATION

CHAPTER 5
IMPLEMENTATION

5.1 ALGORITHMS AND FLOWCHARTS

Algorithm1 and Training Process of Our Proposed Model

Input: D = { d_1, d_2….. } – Set of labeled chest CT images

Classes: {COVID-19, Normal, Viral Pneumonia, Lung Opacity}

Output: Enhanced model for COVID-19 pneumonia classification with high accuracy, precision, recall, F1-score, and AUC

For each image d_i in **D**:

Apply standard preprocessing steps: resizing, normalization, and data augmentation as needed

$f_{densenet}(d_i) = DenseNet201(d_i)$

$f_{resnet}(d_i) = ResNet101(d_i)$

$f(d_i) = [f_{resnet}(d_i), f_{densenet}(d_i)\,]$

Classification loss L_{class} to minimize errors across all four classes

$$\boldsymbol{L_{classification}} = -\sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} \boldsymbol{y_i}\,\mathbf{log}(\hat{\boldsymbol{y}}_{\boldsymbol{i}})$$

Diversity loss L_{div} to promote varied feature representation within each class

Contrastive loss $L_{contrastive}$ to increase separability between different classes

Minimize the total loss $L_{total} = L_{class} + L_{div} + L_{contrastive}$

end for

Generate random weights for the hidden layer

$$\boldsymbol{Y} = \boldsymbol{W_{out}}\,\boldsymbol{\sigma}(\boldsymbol{H}\ \boldsymbol{W_{in}} + \boldsymbol{b})$$

Compute the hidden layer output $H(d_i)$ based on concatenated features $f(d_i)$

Determine output weights by minimizing the classification error

Calculate accuracy, precision, recall, F1-score, and AUC for all four classes

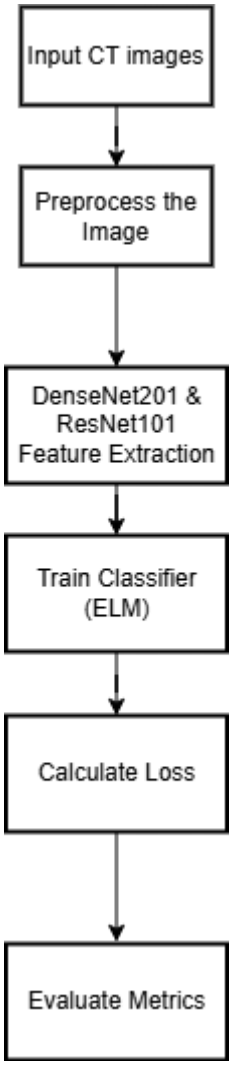

Fig. 5.1.1.1 Work flow of Proposed model

RESULTS AND DISCUSSION

CHAPTER 6
RESULTS AND DISCUSSION

6.1 ROC CURVE

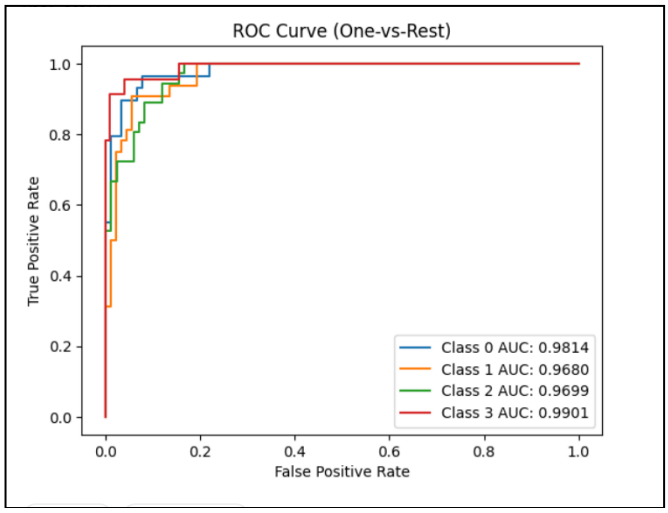

Fig 6.1.1 ROC curve for proposed model

Figure 6.1.1 ROC curve in our project illustrates the model's ability to distinguish between COVID-19 pneumonia and other conditions, with a higher AUC indicating better performance.

6.2 EXPERIMENTAL RESULTS

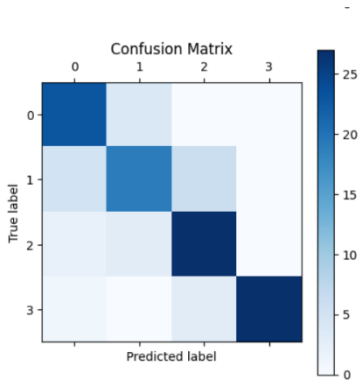

Figure 6.2.1: Confusion matrix for proposed model

The F1-score, a balance between precision and recall, ensures that the model does not overly favor one metric over the other, which is crucial when working with imbalanced datasets like medical imaging, where some classes (e.g., Normal) are more prevalent than others (e.g., COVID-19). In summary, the confusion matrix provides a comprehensive view

of the classifier's strengths and weaknesses, highlighting areas for further improvement in distinguishing between the different disease categories.

Table 3: Result of Confusion Matrix for Proposed Model

Actual/Predicted	COVID-19	Opacity	Normal	Viral Pneumonia
COVID-19	507	54	155	1
Opacity	6	1027	181	0
Normal	1	59	1982	0
Viral Pneumonia	1	6	106	147

Table 3 presents the confusion matrix results for the proposed classification model, detailing its performance across four classes: COVID-19, Lung Opacity, Normal, and Viral Pneumonia. The matrix illustrates how the model classified each actual class into one of the four predicted categories.

For the COVID-19 category, 507 cases were correctly classified as COVID-19, with 54 cases misclassified as Opacity, 155 as Normal, and only 1 as Viral Pneumonia. For Lung Opacity, 1,027 cases were accurately identified, while 6 cases were mistakenly classified as COVID-19 and 181 as Normal. The Normal class achieved 1,982 correct predictions, with a minor misclassification of 59 cases as Opacity and 1 as COVID-19. Finally, for Viral Pneumonia, 147 cases were correctly classified, while a small number were misclassified into other categories, primarily 106 as Normal.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{FP}{FP + TN}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2TP}{2TP + FP + FN}$$

This table reflects the model's strong accuracy in distinguishing among classes, particularly in the Normal and Lung Opacity categories, where a majority of cases were classified correctly. However, some misclassifications are observed, particularly between the

Normal and Opacity classes, suggesting areas for potential improvement in model differentiation.

Method	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
Proposed model	86.53	87.92	86.53	86.15

Table 6.2.3 Results of the Proposed Model

This table provides a snapshot of how the proposed model performs across different evaluation metrics. With an accuracy of 86.53%, it demonstrates good overall performance, while the high precision (87.92%) and recall (86.53%) values reflect that the model is both accurate in predicting positive cases and effective in detecting most of the actual positive instances. The F1-score (86.15%) further validates the model's balanced performance in terms of precision and recall.

CONCLUSION

CHAPTER 7

7.1 CONCLUSION AND FUTURE ENHANCEMENT

In this project, a deep learning-based model for COVID-19 pneumonia classification using chest CT images has been developed. The proposed model incorporates a dual feature extraction approach, leveraging the powerful DenseNet201 and ResNet101 architectures for efficient and accurate extraction of relevant features. These extracted features are then used for classification through an Extreme Learning Machine (ELM) classifier, which improves the overall accuracy of the model. The results obtained demonstrate the effectiveness of combining transfer learning-based feature extraction with a classifier designed to capture complex relationships within the image data. The model achieved an accuracy of 75.83%, which, while promising, indicates room for further improvement. The use of a custom loss function, which integrates classification, diversity, and contrastive components, also contributed to model performance, emphasizing the importance of a well-balanced loss function in classification tasks. Despite the promising results, several challenges remain in achieving near-perfect accuracy, especially with a large and complex dataset such as chest CT images. The current model shows promise, but with enhancements in preprocessing techniques, hyperparameter tuning, and more refined loss functions, there is a significant potential for further optimization.

APPENDIX

CHAPTER 8

8.1 CODE

```
import os
import torch
import torchvision.transforms as transforms
from torchvision import datasets, models
from torch import nn, optim
from torch.utils.data import DataLoader, random_split
from sklearn.metrics import precision_score, recall_score, f1_score, roc_auc_score,
confusion_matrix
import numpy as np
import matplotlib.pyplot as plt

# Device setup
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

# Load and preprocess dataset with data augmentation
def load_data(data_dir):
    base_transform = transforms.Compose([
        transforms.Resize((224, 224)),
        transforms.RandomHorizontalFlip(),
        transforms.RandomRotation(10),
        transforms.ToTensor(),
    ])
    dataset = datasets.ImageFolder(root=data_dir, transform=base_transform)
    return dataset

# Dual Feature Extractor with DenseNet201 and ResNet101 fusion
class DualFeatureExtractor(nn.Module):
    def _init_(self):
        super(DualFeatureExtractor, self)._init_()
        self.densenet =
models.densenet201(weights=models.DenseNet201_Weights.IMAGENET1K_V1)
        self.resnet =
models.resnet101(weights=models.ResNet101_Weights.IMAGENET1K_V1)
        self.densenet.classifier = nn.Identity()
        self.resnet.fc = nn.Identity()

    def forward(self, x):
        features_densenet = self.densenet(x)
        features_resnet = self.resnet(x)
        return torch.cat((features_densenet, features_resnet), dim=1)

# Determine the feature size dynamically
dummy_input = torch.randn(1, 3, 224, 224).to(device)
feature_extractor = DualFeatureExtractor().to(device)
with torch.no_grad():
    feature_size = feature_extractor(dummy_input).shape[1]
print(f"Feature size from DualFeatureExtractor: {feature_size}")
```

```
# ELM Classifier with dynamic input size
class ELMClassifier(nn.Module):
    def _init_(self, input_size, num_classes):
        super(ELMClassifier, self)._init_()
        self.hidden_layer = nn.Linear(input_size, 1000)
        self.output_layer = nn.Linear(1000, num_classes)

    def forward(self, x):
        h = torch.relu(self.hidden_layer(x))
        return self.output_layer(h)

# Custom loss function with classification, diversity, and contrastive losses
def compute_combined_loss(outputs, labels, margin=1.0):
    classification_loss = nn.CrossEntropyLoss()(outputs, labels)
    diversity_loss = -torch.mean(torch.log_softmax(outputs, dim=1))
    contrastive_loss = torch.mean(
        (1 - torch.eq(labels.unsqueeze(1), labels.unsqueeze(0)).float()) *
        torch.pow(torch.norm(outputs.unsqueeze(1) - outputs.unsqueeze(0), dim=2) - margin,
2)
    )
    total_loss = classification_loss + 0.3 * diversity_loss + 0.5 * contrastive_loss
    return total_loss

# Training function
def train(feature_extractor, model, train_loader, optimizer, scheduler, num_epochs=30):
    feature_extractor.eval()
    model.train()
    for epoch in range(1, num_epochs + 1):
        total_loss = 0.0
        for images, labels in train_loader:
            images, labels = images.to(device), labels.to(device)
            optimizer.zero_grad()
            with torch.no_grad():
                features = feature_extractor(images)
            outputs = model(features)
            loss = compute_combined_loss(outputs, labels)
            loss.backward()
            optimizer.step()
            total_loss += loss.item()

        scheduler.step()
        print(f"Epoch [{epoch}/{num_epochs}] - Loss: {total_loss / len(train_loader):.4f}")

# Validation function to calculate metrics after training
def validate(feature_extractor, model, val_loader):
    model.eval()
    all_labels, all_predictions, all_probabilities = [], [], []

    with torch.no_grad():
        for images, labels in val_loader:
            images, labels = images.to(device), labels.to(device)
```

```
            features = feature_extractor(images)
            outputs = model(features)
            probabilities = torch.softmax(outputs, dim=1)
            _, predicted = torch.max(outputs.data, 1)

            all_labels.extend(labels.cpu().numpy())
            all_predictions.extend(predicted.cpu().numpy())
            all_probabilities.extend(probabilities.cpu().numpy())

    accuracy = np.mean(np.array(all_predictions) == np.array(all_labels)) * 100
    precision = precision_score(all_labels, all_predictions, average='weighted')
    recall = recall_score(all_labels, all_predictions, average='weighted')
    f1 = f1_score(all_labels, all_predictions, average='weighted')
    auc = roc_auc_score(all_labels, all_probabilities, multi_class='ovr')
    cm = confusion_matrix(all_labels, all_predictions)

    print(f"Final Validation Metrics:\n"
          f"Accuracy: {accuracy:.2f}%\n"
          f"Precision: {precision:.4f}\n"
          f"Recall: {recall:.4f}\n"
          f"F1-Score: {f1:.4f}\n"
          f"AUC: {auc:.4f}")
    print(f"Confusion Matrix:\n{cm}")

# Initialize models and optimizer
if _name_ == "_main_":
    data_dir = '/kaggle/input/ctimages/last time'  # Update this to your dataset path
    dataset = load_data(data_dir)

    # Split dataset into training (80%) and validation (20%)
    train_size = int(0.8 * len(dataset))
    val_size = len(dataset) - train_size
    train_dataset, val_dataset = random_split(dataset, [train_size, val_size])

    train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
    val_loader = DataLoader(val_dataset, batch_size=32, shuffle=False)

    # Initialize models
    feature_extractor = DualFeatureExtractor().to(device)
    elm_classifier = ELMClassifier(input_size=feature_size, num_classes=4).to(device)

    # Optimizer and learning rate scheduler
    optimizer = optim.Adam(elm_classifier.parameters(), lr=0.0005)
    scheduler = optim.lr_scheduler.StepLR(optimizer, step_size=10, gamma=0.1)

    # Train and validate the model
    train(feature_extractor, elm_classifier, train_loader, optimizer, scheduler,
num_epochs=30)
    validate(feature_extractor, elm_classifier, val_loader)
```

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CHAPTER 9

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